Table of Contents

Purpose/Policy:.....1

Scope:.....2

Definitions/Acronyms:.....2

Authorities/Responsibilities:.....2

Resources and Materials:.....2

PPE Required:.....3

Procedures:.....4

Setting up Nitrogen:.....4

Hooking Up Nitrogen to the BBT:.....7

Testing a Keg for Nitrogen:.....8

Forms:.....8

Reference Documents:.....8

Common Issues:.....8

Document History and Approval:.....9

Purpose/Policy:

The purpose of the “Nitrogen” standard operating procedure is to assist with setting up and how to run kegs with Nitrogen instead of CO2. This SOP will also help educate on the different parts of a nitrogen tank and how to use a liquid Nitrogen dewar.

Scope:

Keg Line operators setting up nitrogen, hooking nitrogen up to the BBT, running kegs with nitrogen instead of CO2, and testing kegs for nitrogen, including use of the liquid nitrogen dewar.

Definitions/Acronyms:

- SOP – Standard Operating Procedure
- PPE – Personal Protective Equipment
- MM – Minomats
- CO2 – Carbon Dioxide
- QC – Quality Control
- BBT – Bright Beer Tank
- PSI – Pounds per Square Inch
- Dewar – Insulated tank that holds liquid nitrogen

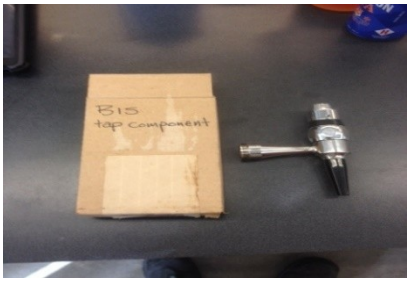
Authorities/Responsibilities:

This SOP can be completed by a lead level packaging operator after proper training from a senior member of the Racking line packaging team. The objective of this SOP is to gain the knowledge needed to operate and use Nitrogen for the keg line and trouble shoot any issues that may arise.

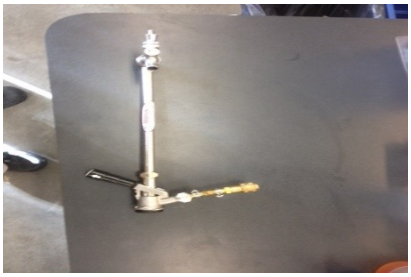
Resources and Materials:

Operations:

- Two 240L Dewar of Liquid Nitrogen
- “Sparkler” tap handle for Nitrogen beers



- “Tower” tap



- C BOX QC
- Small flat head screw driver

Troubleshooting:

- No Equipment required.

PPE Required:

- Hearing protection is suggested for certain steps of this SOP.

Procedures:

Setting up Nitrogen:

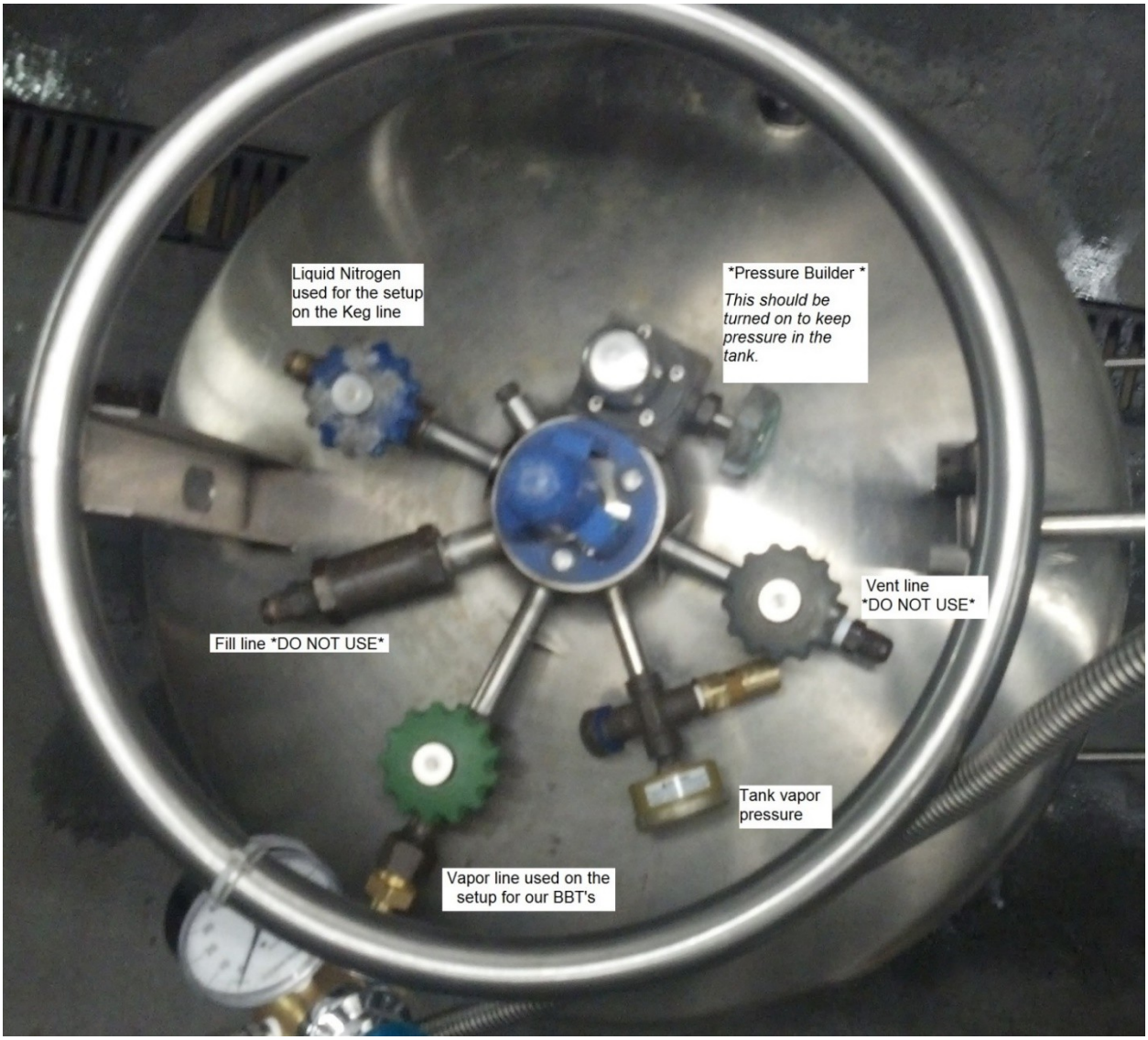


Figure 1.

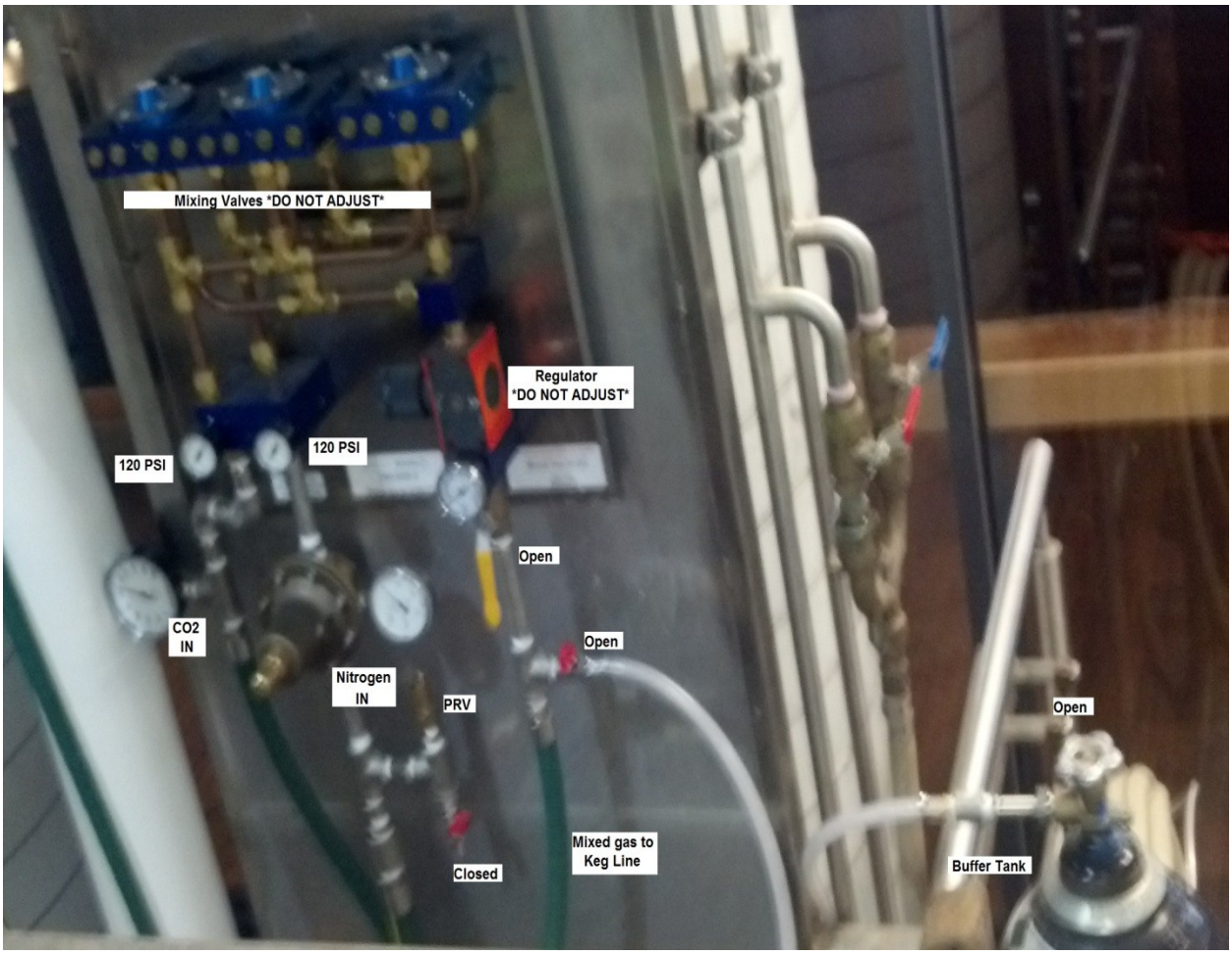


Figure 2.

1. On the day that a Nitrogen beer is to be run make sure the pressure builder on the liquid nitrogen dewar is on. See figure 1.
2. Connect the cryogenic hose between the dewar and the vaporizer located between the keg and can line upstairs. This will supply Nitrogen to the mixing station.
3. On the Nitrogen dewar downstairs between the keg line and can line hook up the inlet to the heat exchanger. Open the liquid Nitrogen supply valve slowly. Make sure the pressure is above 100psi, after 10 minutes once the pressure builder is open.
   3.1 Note: Some dewars are low pressure setups.
4. Turn off the CO2 inlet but make sure beer hall's line is still on (CO2 Growler Machine).

5. Next turn on the CO2 going to the mixing station. Once the mixing station sees the correct pressure on the CO2 and Nitrogen lines it will supply 75% Nitrogen and 25% CO2 blend to the keg line.
6. Go upstairs to the Nitrogen manifold. Open the red Nitrogen bleeder valve and bleed out the line for about a minute. It is highly suggested to wear hearing protection for this step.
7. On the Nitrogen manifold open the mixed gas outlet valve (the valve with the yellow handle).
8. Open the mixed gas (MG) valve on the CO2 hard line.
9. Next open the overflow valve located on the white hose connecting to the mixed gas side arm buffer tank.
10. Remove the false kegs from MM1 and MM2 station 8.
11. Open up the control panels on MM1 and MM2.
12. Inside the panel, look for the solenoid valve with the yellow CO2 labels with the letter “F” and “7” above the solenoid.
13. Using a small flat head screw driver actuate the solenoid valve by turning the yellow screw ½ turn clockwise. This will purge the stations with CO2.
14. Do this step for both MM1”F”, station 7, MM2 “F”, station 7 for 30 seconds each.
15. Shut off manual purge.

Hooking Up Nitrogen to the BBT:



1. Hook up the Nitrogen hose to the vent arm of the tank. Make sure the vent arm stays closed for now.
2. Open the gas valve on the Nitrogen dewar attached to the PSI regulator.
3. Open the PSI regulator to 90+ PSI.
4. Crack the red valve on the top of the stainless cylinder to bleed out any O2.

5. Close the Nitrogen outlet regulator, setting PSI gauge to 0.
6. Next Open the Nitrogen Valve.
7. Open the Nitrogen outlet regulator and set the PSI gauge to 20 PSI.
8. Now open up the vent arm allowing Nitrogen to be used for head pressure.
   8.1 *Note: Make sure the tank's head pressure does not exceed 20 PSI.*
9. Head pressure can be adjusted using the vent arm handle.

Testing a Keg for Nitrogen:

1. Turn the Nitrogen gas to 35 PSI.
2. Hook up the C BOXQC meter to a keg.
3. Hit the rinse button on the meter.
4. Stop the rinse cycle after about 2 minutes.
5. Press the start button on the meter. This will start the measurement and then give you the readings after a short time.

Forms:

Not Applicable

Reference Documents:

Not Applicable

Common Issues:

Not Applicable

Document History and Approval:

Revise document history as described in the table and key below.

(Reference to following document: B1C.SOP.P001: Document Control and Approval Procedure Template)

VERSION NO.	VERSION DATE	DESCRIPTION OF CHANGE	AUTHOR NAME/PROJECT MANAGER
001	2024/08/01		Kyle Szymanski
002	2026/10/26	Updated SOP to Newest Layout Version	Kyle Szymanski

Version Approval Number: 002

Approved by: Kyle Szymanski

Position Title: OSCI Manager

Date: 10/26/26